

量子 2014

一、(1) n 级, n^2

(2) 不随时间改变的力学量且与 H 对易.

$$\frac{d\bar{A}}{dt} = \frac{1}{i\hbar} [A, H] = 0$$

守恒量的本征值及其概率分布不随时间改变.

(3) 受激辐射即处于激发态的发光原子在
外来辐射场的作用下, 向低能态或基态
跃迁时辐射光子的现象. 外来辐射的能量
必须恰好是原子两能级的能量差.

光电效应指在高于极限频率的电磁波照
射下, 某些物质内部的电子吸收能量后逸出
而形成电流.

(4) 一级修正 $H'_{nn} = \langle n | H' | n \rangle$

$$= \text{级} \quad E_n^{(1)} = \sum_{m \neq n} \frac{|\langle m | H' | n \rangle|^2}{E_n^{(0)} - E_m^{(0)}}$$

$$E_n = E_n^{(0)} + \langle n | H' | n \rangle + \sum_{m \neq n} \frac{|\langle m | H' | n \rangle|^2}{E_n^{(0)} - E_m^{(0)}}$$

(5) $\hat{p} = -i\hbar \nabla$

$$L_x = y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} = \frac{i}{\hbar} (y p_z - z p_y)$$

$$L_y = \frac{i}{\hbar} (z p_x - x p_z)$$

$$L_z = \frac{i}{\hbar} (x p_y - y p_x)$$

$$[L_x, L_y] = \left(\frac{i}{\hbar}\right)^2 [y p_z - z p_y, z p_x - x p_z]$$

$$= -\frac{1}{\hbar^2} [y p_z, z p_x] - \frac{1}{\hbar^2} [z p_y, x p_z]$$

$$= -\frac{1}{\hbar^2} y [p_z, z] p_x - \frac{1}{\hbar^2} x [z, p_y] p_z$$

$$= -\frac{i\hbar}{\hbar^2} (x p_y - y p_x) = -L_z.$$

$$\text{同理, } [L_y, L_z] = -L_x$$

$$[L_z, L_x] = -L_y.$$

$$\text{二. } \psi(r, \theta, \varphi) = R(r) Y_{lm}(\theta, \varphi) = \frac{u(r)}{r} Y_{lm}(\theta, \varphi)$$

径向方程满足

$$\frac{d^2}{dr^2} u(r) + \frac{2m}{\hbar^2} \left[E - V(r) + \frac{\hbar^2 l(l+1)}{2mr^2} \right] u(r) = 0$$

$$\frac{d^2}{dr^2} u(r) + \frac{2m}{\hbar^2} \left[E + \frac{e^2}{r} + \frac{e^2}{r^2} + \frac{\hbar^2 l(l+1)}{2mr^2} \right] u(r) = 0$$

$$\text{令 } l'(l'+1) = l(l+1) + \frac{2me^2}{\hbar^2} \quad (1)$$

则

$$\frac{d^2}{dr^2} u(r) + \frac{2m}{\hbar^2} \left[E + \frac{e^2}{r} + \frac{\hbar^2 l'(l'+1)}{2mr^2} \right] u(r) = 0$$

同氢原子方程相似

$$\text{氢原子基态, } E_n = -\frac{e^2}{2a(n_r + l + 1)^2}$$

作变换 $l \rightarrow l'$

$$E = -\frac{e^2}{2a(n_r + l' + 1)^2}, \quad n_r = 0, 1, 2, \dots$$

由 (1) 解得

$$l' = -\frac{1}{2} + (l + \frac{1}{2}) \sqrt{1 + \frac{2me^2}{\hbar^2(l + \frac{1}{2})^2}}$$

$$l, n_r = 0, 1, 2, \dots$$

$$\text{基态, } n_r = l = 0, \text{ 有 } l' = -\frac{1}{2} + \frac{1}{2} \sqrt{1 + \frac{8me^2}{\hbar^2}}$$

$$E = -\frac{e^2}{2a(l'+1)^2}$$

$$= -\frac{2e^2}{a\left(1 + \sqrt{1 + \frac{8me^2}{\hbar^2}}\right)^2}$$

三、同 2017. 第二题.

四、将 $V(r) = \frac{Q}{r} e^{-r/a}$ 代入

$$\sigma(\theta) = \frac{4m^2}{\hbar^4} \left| \int_0^\infty \frac{r \sin(Kr)}{k} \cdot \frac{Q}{r} e^{-r/a} dr \right|^2$$

$$= \frac{4m^2 |Q|^2}{\hbar^4 k^2} \left| \int_0^\infty \sin(Kr) e^{-r/a} dr \right|^2$$

$$I = \int_0^\infty \sin(Kr) e^{-r/a} dr$$

$$= \int_0^\infty -\frac{1}{K} e^{-r/a} d\cos(Kr)$$

$$= -\frac{1}{K} e^{-r/a} \cos(Kr) \Big|_0^\infty - \frac{1}{Ka} \int_0^\infty e^{-r/a} \cos(Kr) dr$$

$$= \frac{1}{K} - \frac{1}{K^2 a} \int_0^\infty e^{-r/a} d\sin(Kr)$$

$$= \frac{1}{K} - \frac{1}{K^2 a} e^{-r/a} \sin(Kr) \Big|_0^\infty - \frac{1}{K^2 a^2} \int_0^\infty e^{-r/a} \sin(Kr) dr$$

$$= \frac{1}{K} - \frac{1}{K^2 a^2} I$$

$$I = \frac{Ka^2}{K^2 a^2 + 1}$$

$$\sigma(\theta) = \frac{4m^2 |Q|^2 a^4}{\hbar^4 (K^2 a^2 + 1)^2}$$

五. (1) 由

$$\begin{vmatrix} E - (A+a) & -b \\ -b & E - (B+a) \end{vmatrix} = 0$$

$$E^2 - (A+B+2a)E + (A+a)(B+a) - b^2 = 0$$

$$E = \frac{A+B+2a}{2} \pm \frac{\sqrt{(A-B)^2 + 4b^2}}{2}$$

(2) 取 $H_0 = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$, $H' = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$.

有 $E_1^{(0)} = A$, $E_2^{(0)} = B$, $\psi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\psi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

一级近似

$$E_1^{(1)} = \langle \psi_1 | H' | \psi_1 \rangle = a$$

$$E_2^{(1)} = \langle \psi_2 | H' | \psi_2 \rangle = a$$

二级近似

$$E_1^{(2)} = \frac{|\langle \psi_2 | H' | \psi_1 \rangle|^2}{E_1^{(0)} - E_2^{(0)}} = \frac{|b|^2}{A-B}$$

$$E_2^{(2)} = \frac{|\langle \psi_1 | H' | \psi_2 \rangle|^2}{E_2^{(0)} - E_1^{(0)}} = \frac{|b|^2}{B-A}$$

$$E_1 = E_1^{(0)} + E_1^{(1)} + E_1^{(2)} = A + a + \frac{|b|^2}{A-B}$$

$$E_2 = E_2^{(0)} + E_2^{(1)} + E_2^{(2)} = B + a + \frac{|b|^2}{B-A}$$

对于精确值有

$$E = \frac{A+B+2a}{2} \pm \frac{A-B}{2} \sqrt{1 + \frac{4b^2}{(A-B)^2}}$$

$$\approx \frac{A+B+2a}{2} \pm \frac{A-B}{2} \left[1 + \frac{1}{2} \frac{4b^2}{(A-B)^2} \right]$$

$$= \left(\frac{A+B}{2} \pm \frac{A-B}{2} \right) + a \pm \frac{|b|^2}{A-B}$$

微扰是精确值的小量近似.

六. (1) 在谐振子势中有

$$E_N = (N + \frac{1}{2}) \hbar \omega, \quad N=0, 1, 2, \dots$$

$$\psi_N(r) = A_N e^{-\alpha^2 r^2/2} H_N(\alpha r), \quad \alpha = \sqrt{\frac{\mu \omega}{\hbar}}$$

自旋 $\frac{1}{2}$ 的全同粒子放入势阱中, 交换反对称.

$$H' = J \hat{S}_1 \cdot \hat{S}_2 = J \frac{1}{2} (\hat{S}^2 - \hat{S}_1^2 - \hat{S}_2^2)$$

$$= \frac{J}{2} (\hat{S}^2 - \frac{3}{2} \hbar^2)$$

H' 本征函数 $|sm_s\rangle$

对于基态,

$$\psi_0 = \psi_0(r_1) \psi_0(r_2) |00\rangle$$

$$E = E_{00} + \langle 00 | H' | 00 \rangle$$

$$= 3\hbar\omega - \frac{3}{4}\hbar^2 J$$

(2) 自旋为 1 的全同粒子

$$H' = \frac{J}{2} (\hat{S}^2 - \hat{S}_1^2 - \hat{S}_2^2) = \frac{J}{2} (\hat{S}^2 - 4\hbar^2), \quad |sm_s\rangle$$

$$\langle sm_s | H' | sm_s \rangle = \frac{J}{2} [s(s+1) - 4] \hbar^2$$

自旋对称态:

$$|00\rangle : -2J\hbar^2$$

$$|22\rangle, |21\rangle, |20\rangle, |2-1\rangle, |2-2\rangle : J\hbar^2$$

自旋反对称态

$$|11\rangle, |10\rangle, |1-1\rangle : -J\hbar^2$$

基态为

$$\psi = \psi_0(r_1) \psi_0(r_2) |00\rangle, \quad E = 3\hbar\omega - 2J\hbar^2$$